

Aditya Kain

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EDUCATION

National Institute of Technology Hamirpur

B.Tech in Electronics and Communication Engineering | CGPA: 7.21/10

Hamirpur, India

Aug. 2023 – Present

CBSE Class XII

Percentage: 90.4%

2021

CBSE Class X

Percentage: 92.6%

2019

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript (ES6+), TypeScript

Frontend: React.js, HTML5, CSS3, Tailwind CSS, D3.js, Socket.io

Backend: Node.js, Express.js, FastAPI, REST APIs, WebSockets, Prisma ORM

Databases: MongoDB, MySQL, ChromaDB

AI / ML: LangChain, RAG Pipelines, Vector Embeddings, Groq API (LLaMA 3), Semantic Search

CS Fundamentals: DSA, OOP, DBMS, Operating Systems, Computer Network, System Design

Developer Tools: Git, GitHub, VS Code, Postman, Vercel, Railway

PROJECTS

RAGChat System | Python, LangChain, ChromaDB, FastAPI, React, Tailwind CSS

May 2026 – Jul. 2026

- Built an **end-to-end RAG pipeline** using **LangChain**, **ChromaDB**, and **Groq (LLaMA 3)** to answer user queries from uploaded PDFs with **context-aware responses** in under **2 seconds**.
- Improved **retrieval accuracy** by implementing **512-token chunking** with 50-token overlap, reducing irrelevant document matches by **~40%** over a naive full-document prompting baseline.
- Developed a **FastAPI** REST backend for **document ingestion** and querying across **5+ simultaneous PDFs** with **API response times below 200ms**.
- Built a **React + Tailwind CSS** chat interface with **source-page citations**; deployed on **Railway** (backend) and **Vercel** (frontend) with **99.9% production uptime**.

Network Intrusion Detection System | Node.js, Express.js, React.js, MySQL

Jan. 2026 – Mar. 2026

- Engineered a **real-time intrusion detection system** processing **1,000+ network packets/sec** with **<50ms** end-to-end latency using **Socket.io** bidirectional WebSockets.
- Applied **graph algorithms** like BFS, Dijkstra, Betweenness Centrality, and Union-Find, boosting **malicious node detection accuracy by 80%** over a sequential-scan baseline across a **50+ node** network.
- Designed an interactive **D3.js dashboard** that reduced **threat identification time by 60%** through real-time network visualization across **50+ nodes**.
- Shipped **10+ REST APIs** using **Express.js** and **Prisma ORM**; optimized **MySQL queries** and indexing to slash **data retrieval latency by 35%** and cut overall **backend response time by 40%**.

ACHIEVEMENTS

- LeetCode:** Solved **300+ problems** and achieved a **contest rating of 1528** (top 25% performer).
- National-Level Hackathon Finalist** at Hack 5.0, selected among the **top teams from 50+ participating teams**.

POSITIONS OF RESPONSIBILITY

Executive Member – Technical & Event Management

Aug. 2023 – Present

Resurgence Club, NIT Hamirpur

- Managed **end-to-end coordination** for a **100+ participant** technical event with zero operational failures.
- Streamlined cross-functional student team workflows, improving execution timelines across **3+ semester-long initiatives**.